

Wormhole Proofs: Recursive Zero-Knowledge Hash-Based Authorization with #P-Hard Privacy

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Abstract. Post-quantum signature schemes such as ML-DSA impose public keys and signatures an order of magnitude larger than their elliptic-curve counterparts, substantially increasing communication and storage costs. An alternative approach replaces signature verification with *hash-based authorization*: a prover demonstrates knowledge of a secret preimage tied to a public commitment, rather than presenting a signature. This sidesteps lattice-based operations entirely, requiring only hash evaluations and membership proofs.

Building on the wormhole mechanism from EIP-7503, we formalize a zero-knowledge *wormhole proof system* where funds are committed to addresses derived from secret preimages and later claimed by proving knowledge of the preimage in zero knowledge. We construct a two-layer recursive proof architecture using Plonky2 with FRI commitments. To achieve privacy against balance-linking adversaries, we introduce client-side dummy padding and local shuffling at the first aggregation layer, enlarging the effective anonymity set.

We give complete game-based security definitions and proofs for deposit binding, one-time withdrawal, spend-path exclusivity, value conservation, nullifier indistinguishability, and transaction unlinkability. Under a uniform prior over deposit subsets consistent with the public view, adversarial identification advantage is bounded by $1/k$ for an anonymity set of size k ; we further prove that even computing k is #P-hard. Our implementation using Plonky2 shows 20 ms proving time per authorization with 1.6 ms verification. Two-layer recursive aggregation scales to 256 authorizations with near-constant final verification (approximately 3.7 ms at the largest batch tested). The marginal on-chain cost is only 56 bytes per settled output, over $80\times$ smaller than a single ML-DSA-87 signature.

Keywords: zero-knowledge proofs · post-quantum cryptography · recursive aggregation · wormhole proofs · anonymity sets · subset sum

1 Introduction

The post-quantum transition poses a fundamental dilemma for blockchains. While Shor’s algorithm breaks elliptic-curve signatures, the standardized post-quantum replacements such as ML-DSA have signatures and public keys an order of magnitude larger than their classical counterparts. Naively substituting them increases on-chain bandwidth and state growth dramatically, without addressing transaction privacy.

Verifying post-quantum signatures inside a zero-knowledge circuit and recursively aggregating them is one path, but the circuit cost remains substantial. This paper takes a different approach: users prove *knowledge of a secret preimage* corresponding to a public wormhole address rather than proving a signature. This idea builds on Ziggy’s hash-preimage signatures and EIP-7503’s wormhole concept, but we develop it into a practical post-quantum private settlement system.

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We formalize the wormhole proof relation, design a two-layer recursive aggregation protocol with client-side privacy mechanisms, prove six core security properties in the game-based style, and show that computing the effective anonymity set is $\#P$ -hard. Our Plonky2 implementation achieves practical performance with only 56 bytes of marginal on-chain cost per settled output.

Our Contributions

- Formalization of the wormhole proof relation and a precise security model for post-quantum private settlement.
- A two-layer recursive aggregation architecture with client-side dummy padding and shuffling that enlarges anonymity sets while remaining efficient.
- Proof that computing the size of the anonymity set is $\#P$ -hard, together with game-based security proofs for deposit binding, one-time withdrawal, spend-path exclusivity, value conservation, nullifier indistinguishability, and transaction unlinkability.
- A practical Plonky2 implementation achieving 20 ms per authorization and 56 bytes marginal on-chain cost per settled output, over $80\times$ smaller than an ML-DSA-87 signature, with near-constant recursive verification (under 4 ms for up to 256 authorizations).

Paper Organization

Section 2 reviews background on wormholes, privacy pools, and transparent recursion with FRI. Section 3 defines our security model. After a protocol overview in Section 4 and formal interfaces in Section 5, Section 6 details the recursive construction. Section 7 contains our security definitions and proofs. We report performance in Section 8 and discuss related work in Section 9.

2 Background and Design Goals

2.1 Wormholes and Privacy Pools

EIP-7503 proposes *zero-knowledge wormholes*: deposit to secret-derived addresses, then prove knowledge of the secret to withdraw [KBK+23]. We extend this concept with post-quantum proving and recursive aggregation with dummy padding for privacy.

Related systems include privacy pools and shielded ledgers like Zcash and Tornado Cash [KBK+23, BIN+23, HBH+23, PSS19]. Unlike fully shielded ledgers, deposits remain visible as ordinary transfer events on-chain; privacy comes from proving knowledge of a secret without revealing which deposit funded the withdrawal. The circuit authenticates deposits via inclusion in a consensus-maintained *ZK tree*—a 4-ary sorted Poseidon Merkle tree of transfer events whose root is committed in the block header—rather than via encrypted note commitments or Merkle–Patricia storage proofs. Importantly, no ciphertexts are stored on-chain: in the standard post-quantum threat model (where Shor’s algorithm breaks DLP- and factoring-based encryption but 256-bit hashes retain large quantum security margins), there is no harvested material that becomes decryptable later. Retroactive privacy rests only on the preimage resistance of the hash function—the same assumption underlying the system’s soundness—rather than on any additional encryption assumption.

2.2 Transparent Recursion

We use Plonky2, combining PLONK arithmetization with FRI commitments for recursive verification without trusted setup [Pol22, BBHR18]. FRI is based on hash functions rather than elliptic curves, making it plausibly post-quantum secure: known quantum attacks on hash preimage search (Grover) and collision finding (BHT) improve over classical bounds, but 256-bit outputs retain comfortable margins at our security level. Poseidon2 provides efficient in-circuit hashing over prime fields [GKS23].

Security Parameters: Our instantiation uses the Goldilocks field ($p = 2^{64} - 2^{32} + 1$) with a quadratic extension for 100-bit conjectured soundness. FRI is configured with 28 query rounds, rate 1/8 (blowup factor 2^3), and 16 bits of proof-of-work grinding, yielding $28 \times 3 + 16 = 100$ bits of conjectured security against proof forgery under Plonky2’s standard FRI soundness conjecture; provable bounds in the Johnson list-decoding regime are lower [Pol22, BBHR18]. Poseidon2 outputs 4 field elements (256 bits), giving roughly 128-bit classical collision resistance and 256-bit classical preimage resistance. Against a quantum adversary, Grover search reduces preimage work to about 2^{128} operations, while generic quantum collision attacks (e.g., BHT) reduce collision work to about 2^{85} operations—still above our 100-bit soundness target. Hash output length, not the Goldilocks field size, therefore governs post-quantum hash margins.

2.3 Design goals

The protocol aims to satisfy four simultaneous goals:

1. **Amortized Verification:** The marginal on-chain cost of verifying an additional withdrawal should be small.
2. **Deposit Binding:** Every withdrawal must be bound to a real deposit recorded in authenticated chain state.
3. **Replay Prevention:** A deposit should not authorize multiple withdrawals; nullifiers must uniquely bind proofs to deposits.
4. **Meaningful Privacy:** The chain should reveal only the public information needed for settlement: exits, amounts, the asset identifier, shared fee parameters, nullifiers, and a block hash with its block number. Funding accounts, secrets, and ZK tree inclusion proofs should remain private.

3 System, Adversary, and Trust Model

3.1 Roles

- **User / Prover:** Generates a secret, derives a wormhole address, deposits funds, and later produces a zero-knowledge proof to withdraw. First-layer aggregation should be performed locally to preserve privacy.
- **Aggregator:** Combines multiple proofs into a single aggregate proof. This role may be the user, a third-party service, or a block producer. Higher-layer aggregation can be delegated since it operates only on public outputs.
- **Blockchain / Verifier:** Validates the final proof, checks nullifier freshness, and settles withdrawals to the specified exit accounts.

3.2 Adversarial capabilities

We assume a probabilistic polynomial-time adversary that can:

- observe the full chain and all public proof outputs,
- submit arbitrary deposits,
- perform timing analysis, and
- attempt to replay or malformed proof submissions.

The adversary cannot:

- break the collision resistance of the hash function,
- break the knowledge soundness of the proof system, or
- violate the chain’s block-history availability.

3.3 Trust assumptions

The protocol relies on the following assumptions.

1. **Hash Security:** The Poseidon2 instantiation used by the system remains collision resistant and preimage resistant for the relevant security level [GKS23].
2. **Proof Soundness:** The configured Plonky2 recursion stack remains knowledge sound under its standard assumptions [Pol22, BBHR18].
3. **Authenticated History:** The blockchain can check that the public block hash and block number correspond to a valid block in chain history.
4. **Witness Availability:** Honest users can obtain the ZK tree inclusion proof needed to build a witness for their deposit.

4 Protocol Overview

The protocol consists of five stages.

1. **Secret Generation and Deposit:** The user samples a secret s and derives a wormhole address

$$\text{WA}(s) = H(H(\text{salt}_{\text{wh}} \parallel s)).$$

Because this address is derived from a hash rather than a public key, no one can spend from it via ordinary signature-based authorization—wormhole addresses can receive funds but never have outgoing transactions. Anyone can deposit to a wormhole address through an ordinary on-chain transfer, creating a record that can later be authenticated inside the circuit. A user may receive funds from their own prior transfers or from third-party payments to wormhole addresses derived from secrets the user controls.

2. **Witness Acquisition:** To withdraw, the user gathers the deposit record (a transfer event to their wormhole address) and a ZK tree inclusion proof showing that event is included under the block’s committed tree root. These form the private witness.

3. Leaf Proof Generation: The user chooses up to two exit outputs, computes a nullifier

$$\text{Null}(s, c) = H(H(\text{salt}_{\text{null}} \parallel s \parallel c)),$$

and produces a leaf proof. Its intermediate public interface contains the nullifier, the exits, the output amounts, the asset identifier, the fee basis points, the block reference, and the authenticated input amount. This interface is consumed only by the user's local first-layer wrapper; leaf proofs are not submitted to the chain. The wrapper's zero-knowledge proof hides the input amounts, secrets, and ZK tree inclusion proofs.

4. First-Layer Aggregation: The user batches leaf proofs that share the same `block_hash`, asset, and fee rate, pads with dummy leaves if needed, shuffles the order, and produces a zero-knowledge wrapper proof that forwards grouped exits and canonically sorted nullifiers. The wrapper enforces the fee inequality once over the sums of all real inputs and outputs in the batch. Each leaf retains one input and two output slots, but an individual leaf's outputs need not fit within that leaf's input; value may be pooled across leaves as long as the private batch is solvent in aggregate. A user may therefore combine multiple deposits from distinct wormhole addresses into one private settlement segment without exposing the deposit-to-output amount assignment. This step should be performed locally to preserve privacy.

5. Settlement: The blockchain verifies the aggregate proof, checks that the public `block_hash` matches the referenced block number, verifies that the fee parameter matches the configured rate, rejects reused nullifiers, and settles the exits. A public batch preserves the boundaries of its constituent private segments. For each accepted segment with total exit S_q , settlement charges $\lceil S_q f / (10000 - f) \rceil$ quantized units and sums those per-segment ceilings; applying one ceiling to the combined public-batch output would not match the rounding boundaries proved by the private segments.

5 Formal Specification and Interfaces

5.1 Leaf public inputs

The leaf proof exposes the following public fields:

- `asset_id`: Token identifier.
- `output_amount_1`, `output_amount_2`: Exit amounts (secondary is zero if unused).
- `fee_bps`: Fee parameter in basis points.
- `nullifier`: Replay-prevention tag derived from secret and counter.
- `exit_account_1`, `exit_account_2`: Destination accounts.
- `block_hash`: Commitment to block header fields.
- `block_number`: Height of the referenced block.
- `input_amount`: Deposited amount before fee deduction, appended to the intermediate leaf interface and hidden by first-layer aggregation.

5.2 Leaf private witness

The private witness includes:

- **secret**: User secret from which nullifier and wormhole address are derived.
- **counter**: Per-deposit counter (**transfer_count**) used in nullifier derivation.
- **wormhole_account**: Derived wormhole address $WA(s)$.
- **zk_tree_proof**: Merkle inclusion proof for the deposit transfer event in the block's ZK tree.
- Header fields (**parent_hash**, **state_root**, **extrinsics_root**, **zk_tree_root**, **digest**) used to recompute and bind the public **block_hash**.

5.3 Leaf relation

Let $x = (a, v_1, v_2, f, n, \alpha_1, \alpha_2, h_B, b, v_{in})$ denote the intermediate public input tuple, with v_{in} appended after the pre-existing fields, and let w denote the witness described above. The leaf relation $\mathcal{R}_{leaf}$ holds when there exists a witness w such that:

- (C1) **Nullifier Correctness**: $n = \text{Null}(s, c)$.
- (C2) **Wormhole-Address Correctness**: $\alpha_{wh} = WA(s)$.
- (C3) **Authenticated Inclusion**: The ZK tree inclusion proof verifies that a transfer event to $WA(s)$ with counter c and amount v_{in} exists under the tree root committed in the block header.
- (C4) **Block Binding**: The inclusion proof is bound to the public block hash h_B and block number b via the header's **zk_tree_root** field and the circuit's block-hash commitment.

The leaf circuit range-checks and authenticates v_{in}, v_1, v_2 , but deliberately imposes no per-leaf fee or conservation inequality. The two output slots remain fixed; conservation is enforced over their sums in the locally generated private-batch proof. In formulas that use basis points, f denotes the integer fee parameter. In the privacy analysis we overload f to denote the fractional rate $f/10^4$, so volume thresholds take the form $S/(1 - f)$.

5.4 On-chain verification

The proof system does not, by itself, settle a withdrawal. The blockchain must also:

- verify the referenced block exists in chain history
- check the fee parameter matches the configured rate
- reject previously-seen nullifiers
- compute and sum the volume-fee ceiling separately for each accepted private segment
- credit exit accounts without increasing total supply

6 Recursive Aggregation Architecture

6.1 First-layer aggregation

The first aggregation layer is privacy-sensitive and should be executed locally. A batch of up to N leaf proofs is aggregated as follows:

- All real leaves must share the same `block_hash`, `asset_id`, and `fee_bps`.
- If fewer than N real leaves are available, the batch is padded with dummy leaves.
- The leaf order is shuffled before wrapping.
- Exits to identical accounts are grouped into a single summed output.
- Real nullifiers are forwarded; dummy nullifiers are replaced by $H(H(u))$ for prover-supplied random preimages u , preventing a malicious prover from choosing a dummy tag that collides with a real one.
- The resulting nullifiers are sorted in canonical digest order, independently of the exit-slot layout.

We call one first-layer output a *private settlement segment*. For its set R of real leaves, define

$$I_q = \sum_{i \in R} v_{\text{in},i} \quad \text{and} \quad S_q = \sum_{i \in R} (v_{i,1} + v_{i,2}).$$

In addition to recursive verification, metadata consistency, exit grouping, and nullifier handling, the private-batch relation enforces

$$S_q \cdot 10000 \leq I_q \cdot (10000 - f).$$

Dummy inputs and outputs are masked to zero before summation. Neither I_q nor any $v_{\text{in},i}$ is forwarded to the aggregate proof. The relation preserves the one-input–two-output leaf shape while allowing cross-leaf value pooling: a leaf may carry outputs exceeding its own input if other real inputs in the same segment cover the difference.

A delegated first-layer service would observe raw child proofs before padding and shuffling, collapsing much of the anonymity set quantified in [section 7.5](#).

6.2 Higher-layer aggregation

Higher layers aggregate already-wrapped proofs. Since they operate only on public outputs, they can be safely delegated, and zero-knowledge is not required in principle at these layers. They verify already fee-valid private segments and preserve segment boundaries; they do not introduce a second fee inequality or merge the segments' integer-rounding boundaries.

7 Security and Privacy Analysis

We formalize security via game-based definitions for the properties a wormhole system must satisfy. Let $\mathcal{B}$ denote a probabilistic polynomial-time adversary, H a hash function, and $\Pi = (\text{Prove}, \text{Verify})$ the proof system with knowledge soundness error ϵ_{ks} .

7.1 Deposit binding

Definition 1 (Deposit Binding Game). The *deposit binding game* $\text{Game}_{\text{bind}}^{\mathcal{B}}$ proceeds as follows:

1. $\mathcal{B}$ is given access to the blockchain state and may create arbitrary deposits.
2. $\mathcal{B}$ outputs a leaf proof π and its intermediate public input tuple x as defined in [section 5](#).
3. $\mathcal{B}$ wins if $\text{Verify}(\pi, x) = 1$ and there is *no* witness $w = (s, c, \dots)$ satisfying $\mathcal{R}_{\text{leaf}}(x, w)$ whose authenticated deposit—the ZK tree leaf recording a transfer to $\text{WA}(s)$ with counter c , asset a , and input amount v_{in} —is actually recorded in the ZK tree at block (h_B, b) .

We say the system satisfies *deposit binding* if $\Pr[\text{Game}_{\text{bind}}^{\mathcal{B}} = 1] \leq \epsilon_{\text{ks}} + \epsilon_{\text{coll}}$ for all PPT $\mathcal{B}$, where ϵ_{coll} is the collision probability of H .

Theorem 1 (Deposit Binding). *The wormhole protocol satisfies deposit binding under knowledge soundness of Π and collision resistance of H .*

Proof. Suppose $\mathcal{B}$ wins. By knowledge soundness, the extractor outputs a witness w satisfying $\mathcal{R}_{\text{leaf}}(x, w)$ —and hence conditions (C1)–(C4)—except with probability ϵ_{ks} . Because $\mathcal{B}$ wins, no satisfying witness has its deposit recorded in the tree; in particular, the extracted witness’s deposit tuple $(\text{WA}(s), c, a, v_{\text{in}})$ is absent from the ZK tree at block (h_B, b) . Yet condition (C3) requires w to include a ZK tree inclusion path for exactly this tuple that verifies against the root committed in (h_B, b) per (C4). A verifying path for an absent leaf requires a collision in the tree’s Poseidon hash (probability ϵ_{coll}). The blockchain’s validation that (h_B, b) refers to a real block ensures the committed root is the correct one for that height. $\square$

7.2 One-time withdrawal

Definition 2 (Double-Spend Game). The *double-spend game* $\text{Game}_{\text{double}}^{\mathcal{B}}$ proceeds as follows:

1. $\mathcal{B}$ creates a single deposit of amount v to address $\text{WA}(s)$ for adversarially chosen s .
2. $\mathcal{B}$ outputs two proofs (π_1, x_1) and (π_2, x_2) with nullifiers $n_1 \neq n_2$.
3. $\mathcal{B}$ wins if both proofs verify and both claim outputs from the same deposit.

We say the system satisfies *one-time withdrawal* if $\Pr[\text{Game}_{\text{double}}^{\mathcal{B}} = 1] \leq \epsilon_{\text{ks}} + \epsilon_{\text{coll}}$, where ϵ_{coll} is the collision probability of H .

Theorem 2 (One-Time Withdrawal). *The wormhole protocol satisfies one-time withdrawal under knowledge soundness of Π and collision resistance of H .*

Proof. Suppose $\mathcal{B}$ wins. By knowledge soundness, there exist witnesses w_1, w_2 with secrets s_1, s_2 and counters c_1, c_2 such that both proofs reference the same deposit. Since both reference the same deposit address, $\text{WA}(s_1) = \text{WA}(s_2)$, which by collision resistance implies $s_1 = s_2$. The counter c is part of the ZK tree leaf and verified by the inclusion proof; since both proofs reference the same deposit, they must use the same counter $c_1 = c_2$. But then $n_1 = \text{Null}(s_1, c_1) = \text{Null}(s_2, c_2) = n_2$, contradicting $n_1 \neq n_2$. Thus $\mathcal{B}$ can only win by breaking knowledge soundness or finding a collision. $\square$

Since the blockchain maintains a nullifier set and rejects duplicates, each deposit can be withdrawn at most once.

7.3 Spend-path exclusivity

Definition 3 (Signature Forgery Game). The *signature forgery game* $\text{Game}_{\text{sig}}^{\mathcal{B}}$ proceeds as follows:

1. Challenger samples $s \leftarrow \{0, 1\}^{256}$ and publishes $\text{WA}(s)$.
2. $\mathcal{B}$ outputs a keypair (sk, pk) and a signature σ on an arbitrary message m .
3. $\mathcal{B}$ wins if $H(pk) = \text{WA}(s)$ and $\text{SigVerify}(pk, m, \sigma) = 1$.

We say the system satisfies *spend-path exclusivity* if $\Pr[\text{Game}_{\text{sig}}^{\mathcal{B}} = 1] \leq \epsilon_{\text{pre}}$, where ϵ_{pre} is the preimage resistance of H .

Theorem 3 (Spend-Path Exclusivity). *The wormhole protocol satisfies spend-path exclusivity under preimage resistance of H .*

Proof. To win, $\mathcal{B}$ must find pk such that $H(pk) = \text{WA}(s) = H(H(\text{salt}_{\text{wh}} \parallel s))$. This requires either:

1. producing the structured inner value itself, $pk = H(\text{salt}_{\text{wh}} \parallel s)$ for the challenge s , as a public key $\mathcal{B}$ controls, or
2. finding some other pk such that $H(pk)$ equals the target address.

Both are preimage attacks on the outer hash. For post-quantum schemes where $|pk| > |H|$ (e.g., ML-DSA), case (1) is impossible since a 256-bit hash output cannot equal a larger public key. In either case, success probability is bounded by ϵ_{pre} . $\square$

7.4 Value conservation

Definition 4 (Inflation Game). The *inflation game* $\text{Game}_{\text{infl}}^{\mathcal{B}}$ proceeds as follows:

1. $\mathcal{B}$ creates deposits totaling value V .
2. $\mathcal{B}$ submits a sequence of valid private segments resulting in withdrawals totaling value W .
3. $\mathcal{B}$ wins if $W \cdot 10000 > V \cdot (10000 - f)$, where f is the common fee rate in basis points.

Theorem 4 (Value Conservation). *The wormhole protocol satisfies value conservation: no adversary can win $\text{Game}_{\text{infl}}$ except with negligible probability.*

Proof. Value conservation follows from three properties:

- Each accepted private segment q enforces $S_q \cdot 10000 \leq I_q \cdot (10000 - f)$ over its real leaves.
- Aggregation preserves output sums; grouping outputs to identical destinations does not change S_q .
- One-time withdrawal (theorem 2) and the blockchain's nullifier check prevent a deposit from contributing to more than one accepted segment.

Thus the segment input sets are disjoint and $\sum_q I_q \leq V$. Summing the segment inequalities gives

$$W \cdot 10000 = \sum_q S_q \cdot 10000 \leq \sum_q I_q (10000 - f) \leq V(10000 - f),$$

contradicting the winning condition.

For $0 \leq f < 10000$, define the on-chain fee for segment q as

$$F_q = \left\lceil \frac{S_q f}{10000 - f} \right\rceil.$$

The proved inequality implies $S_q + F_q \leq I_q$, so user exits plus the fee distributed or burned by settlement remain covered by that segment's deposits. A public batch must therefore charge $\sum_q F_q$ over its accepted private segments. Charging $\left\lceil (\sum_q S_q) f / (10000 - f) \right\rceil$ once can be smaller because of integer rounding and does not preserve the exact fee boundary attested by each private segment. $\square$

7.5 Anonymity sets

An adversary observing a first-layer aggregate sees N canonically sorted nullifiers (some real, some dummies), grouped exits, and the public metadata (b, a, f) . Deposit unlinkability is analyzed by enumerating plausible funding subsets in $\mathcal{D}(b, a)$; nullifier indistinguishability is a separate property that hides padding and batch composition. Throughout this subsection, H denotes the Poseidon2 hash used in-circuit, modeled as a random oracle with 256-bit outputs.

Dummy nullifiers. Real nullifiers are $\text{Null}(s, c) = H(H(\text{salt}_{\text{null}} \parallel s \parallel c))$ as in section 5. Dummy padding replaces each dummy slot with

$$\text{DNull}(u) = H(H(u)),$$

where the prover samples $u \leftarrow \{0, 1\}^{256}$ and the aggregation circuit recomputes $H(H(u))$ from the witnessed preimage (matching the implementation).

Definition 5 (Nullifier Indistinguishability Game). The *nullifier indistinguishability game* $\text{Game}_{\text{null}}^{\mathcal{B}}$ proceeds as follows:

1. Challenger samples secret $s \leftarrow \{0, 1\}^{256}$, counter $c \in \mathbb{N}$, and dummy preimage $u \leftarrow \{0, 1\}^{256}$.
2. Challenger computes $n_0 = \text{Null}(s, c)$ and $n_1 = \text{DNull}(u)$.
3. Challenger flips $b \leftarrow \{0, 1\}$ and sends n_b to $\mathcal{B}$.
4. $\mathcal{B}$ outputs guess b' .
5. $\mathcal{B}$ wins if $b' = b$.

We say nullifiers are *indistinguishable* if for all PPT $\mathcal{B}$ making at most q_H queries to the random oracle H ,

$$\left| \Pr[\text{Game}_{\text{null}}^{\mathcal{B}} = 1] - \frac{1}{2} \right| \leq \epsilon_{\text{ro}}(q_H),$$

where $\epsilon_{\text{ro}}(q_H)$ is negligible in the hash output length $\lambda = 256$ for fixed q_H .

Theorem 5 (Nullifier Indistinguishability). *In the random oracle model, real and dummy nullifiers are computationally indistinguishable: $\epsilon_{\text{ro}}(q_H) = O(q_H/2^\lambda)$.*

Proof. Write $p = \text{salt}_{\text{null}} \parallel s \parallel c$ and let $x_0 = H(p)$, $x_1 = H(u)$, so $n_0 = H(x_0)$ and $n_1 = H(x_1)$. The fixed salt and public counter c do not weaken the analysis: s is uniform and independent of u , so p is unpredictable to $\mathcal{B}$ before the challenge unless it queries $H(p)$.

In the random oracle model, x_0 and x_1 are uniform and independent on their first queries, and so are n_0 and n_1 unless $\mathcal{B}$ has previously queried $H(x_0)$ or $H(x_1)$. Because s

is uniform and unknown to $\mathcal{B}$, the real preimage p is unpredictable (even though $\text{salt}_{\text{null}}$ and c are fixed or public), and the dummy preimage u is uniform and independent. The probability that any of $\mathcal{B}$'s q_H queries hits p , u , x_0 , or x_1 before the challenge is therefore at most $4q_H/2^\lambda$ by a union bound over the four unpredictable points. Conditioned on no such query, n_0 and n_1 are identically distributed and independent of b , so $\Pr[b' = b] = 1/2$. $\square$

Instantiating H with Poseidon2 is heuristic, as is standard for random-oracle-based designs: collision and preimage resistance alone do not imply the pseudorandomness used above, but no structural property of Poseidon2 is known that would distinguish double-hash outputs of unpredictable inputs.

Definition 6 (Aggregate view). A first-layer aggregate Y exposes block number b , asset identifier a , fee rate f , batch capacity N , a canonically sorted list of N nullifiers, and grouped exits $\{(\alpha_j, e_j)\}_{j=1}^E$ with total $S = \sum_{j=1}^E e_j$. For candidate-set analysis the sorted nullifier list is equivalently treated as a multiset.

Definition 7 (Plausible deposit subsets). Let $\mathcal{D}(b, a)$ be the unspent wormhole deposits recorded in the ZK tree at block b with asset a , with amounts $a_1, \dots, a_M$. A subset $I \subseteq \{1, \dots, M\}$ is *plausible* for Y if:

- (i) **Batch size:** $|I| \leq N$.
- (ii) **Output slots:** $E \leq 2|I|$, since each authorization exposes at most two outputs before grouping.
- (iii) **Volume:** $\sum_{i \in I} a_i \geq S/(1-f)$ (equivalently, $\sum_{i \in I} a_i(1-f) \geq S$).

The *effective anonymity set* is

$$\mathcal{A}(Y) = \{I \subseteq \{1, \dots, M\} \mid I \text{ is plausible for } Y\}.$$

These are the amount and cardinality constraints relevant to funding-subset feasibility. There is no longer a cap $\sum_j x_{ij} \leq a_i(1-f)$ tying the two output slots of leaf i to that leaf's input: only the segment-wide volume inequality and the $2|I|$ available slots apply. For example, at zero fee, deposits $\{100, 2\}$ can explain three exits of 34 each even though one raw leaf must carry an output larger than its own input; the aggregate totals are both 102, and three exits fit in four slots. The true funding subset lies in $\mathcal{A}(Y)$. The definition depends on exits and metadata only; the aggregate hides the intermediate input amounts, and published nullifiers do not identify deposits without inverting $\text{Null}(s, c)$.

Theorem 6 (Transaction Unlinkability). *Let Y be a first-layer aggregate with $|\mathcal{A}(Y)| = k \geq 1$. Assume the true funding subset is drawn uniformly from $\mathcal{A}(Y)$. Then no PPT adversary can identify the true subset with probability greater than $1/k$.*

Proof. An adversary attempting to identify the funding deposits observes Y and the deposit pool $\mathcal{D}(b, a)$, computes $\mathcal{A}(Y)$, and guesses which subset was used. Any guess outside $\mathcal{A}(Y)$ is inconsistent with the public view—it violates the batch cap, the two-output slot bound, the block/asset filter, or the aggregate volume threshold—and can be ruled out without guessing.

Under the uniform prior over $\mathcal{A}(Y)$, every plausible subset is equally likely, so success probability is at most $1/k$. Any strategy exceeding $1/k$ must use information outside this model (e.g., side channels, deposit clustering, or a non-uniform prior over user behavior). Nullifiers are not needed for this attack: each published nullifier is a one-way tag $\text{Null}(s, c)$ that does not reveal which deposit funded the withdrawal. $\square$

Role of nullifier indistinguishability. *Theorem 5* addresses a different question: whether an adversary can tell real nullifiers from dummy padding, or infer how many real authorizations appear in a batch. This hides batch composition but does not change the deposit-unlinkability analysis above. Without indistinguishability, an adversary who learns the number of real authorizations could restrict attention to subsets of that exact size, potentially shrinking the effective candidate set below k .

The implemented layout does not expose a positional nullifier-to-exit association. After dummy replacement, the private-batch circuit sorts all nullifier digests into canonical order, while exits are grouped in a separate region. A non-zero exit slot therefore cannot mark any particular sorted nullifier as real. The exits alone reveal the unavoidable lower bound $\lceil E/2 \rceil$ on the number of real leaves, because each leaf has two slots; the nullifier ordering adds no slot-specific leakage.

Because the private-batch relation enforces an inequality rather than exact equality, the adversary must consider any subset whose aggregate deposits can fund at least S after fees. A user may voluntarily under-withdraw by choosing S below the segment maximum, enlarging the volume-feasible candidate set; after the chain charges the fee derived from S , the remaining deposit value stays locked at the unspendable addresses and is effectively burned. Beyond aggregate volume, the binding structural constraint is the unchanged two-output slot count, not a public assignment of output amounts to individual deposits.

Complexity of counting plausible subsets: The decision version of subset sum is NP-complete [GJ79]. The counting version $\#SUBSETSUM$ is $\#P$ -complete [Val79]: the standard reduction establishing NP-completeness of subset sum is parsimonious, so it transfers to the counting class. In the special case $E = 1$ and $N = M$, condition (ii) is vacuous and $\mathcal{A}(Y)$ reduces to threshold $\#SUBSETSUM$ with $T = S/(1 - f)$.

Theorem 7 (Hardness of Anonymity Set Computation). *Computing $|\mathcal{A}(Y)|$ is $\#P$ -hard.*

Proof. We reduce from $\#SUBSETSUM$. Given an instance $(a_1, \dots, a_M, T)$, construct an aggregate view with fee rate $f = 0$ (avoiding any quantization of amounts), $E = 1$, exit (α_1, e_1) where $e_1 = T$, batch cap $N = M$, and deposit pool $\mathcal{D}(b, a)$ containing the same amounts. For any $I \subseteq \{1, \dots, M\}$, condition (iii) holds iff $\sum_{i \in I} a_i \geq T$, and condition (ii) holds because $E = 1 \leq 2|I|$ for every non-empty I . Thus $|\mathcal{A}(Y)|$ equals the number of subsets summing to at least T . Given oracle access to this count, two calls with thresholds T and $T + 1$ recover $\#SUBSETSUM$ exactly as in the standard threshold-to-exact conversion. Hence computing $|\mathcal{A}(Y)|$ is $\#P$ -hard. $\square$

Amount-linkage improvement. Under per-leaf conservation, an observer could reject candidate subsets whose individual deposit caps could not be routed to the published per-account totals. Aggregate conservation removes that pruning signal: the vector of authenticated inputs is visible only inside the local wrapper, and no output slot is publicly funded by a corresponding input. Candidate subsets are therefore filtered by total volume and slot count rather than a deposit-by-deposit transportation problem. This does not hide deposits or exits themselves, but it removes a deterministic amount-linkage channel between them.

Burn Cost and Bounded Rationality: A rational adversary may assume users are unwilling to leave significant value locked for privacy. Under bounded rationality, the effective anonymity set shrinks to plausible subsets whose total deposit volume is at most $(1 + \delta) \cdot S/(1 - f)$ (e.g., 1%)—that is, at most slightly above the minimum volume required by condition (iii).

Table 1: First-layer aggregation times for N leaf proofs.

N leaf proofs	Prove (s)	Verify (ms)
2	1.55	2.75
4	2.81	2.88
8	5.39	3.02
16	10.74	3.16
32	21.71	3.55

Table 2: Second-layer aggregation times. Each first-layer proof aggregates 8 leaf proofs.

M first-layer proofs	Total leaves	Prove (s)	Verify (ms)
2	16	0.92	2.72
4	32	1.88	2.86
8	64	3.84	3.03
16	128	7.97	3.24
32	256	16.69	3.74

7.6 Privacy leakage

Output amounts, exit addresses, nullifiers, and timing are public. The unlinkability bound in [theorem 6](#) assumes a uniform prior over the plausible deposit subsets $\mathcal{A}(Y)$; behavioral amount patterns (e.g., round numbers), address reuse, deposit clustering, and timing correlation can shrink the effective set further. Aggregate enforcement hides the input-amount vector and its assignment to outputs, but the total-volume threshold remains inferable from S and f . Nullifier indistinguishability ([theorem 5](#)) hides dummy padding but is orthogonal to deposit identification. First-layer padding and cross-leaf pooling enlarge plausible deniability, not full shielding.

8 Evaluation

8.1 Methodology

Benchmarks were executed on an Apple M2 Max with 12 CPU cores and 32 GB of RAM. Circuit artifacts were pre-generated; reported times measure only proving and verification. Prove times include witness assignment and proof generation. Verify times measure verification of a pre-generated proof.

All first-layer measurements use repeated identical dummy leaf proofs with the same `block_hash`. Second-layer benchmarks use repeated first-layer proofs.

8.2 Leaf proofs

Individual leaf proof generation takes approximately 20 ms, with verification in 1.6 ms. These times are dominated by proving; witness assignment is negligible.

8.3 First-layer aggregation

Proving time scales with batch size, while verification remains in the millisecond range across all tested configurations.

8.4 Second-layer aggregation

Second-layer verification time grows slowly with batch size (2.72 ms at 16 leaves to 3.74 ms at 256 leaves), remaining in the low millisecond range and demonstrating the benefit of recursive composition.

8.5 On-chain footprint

The final aggregate proof is constant-size regardless of the number of authorizations. The appended leaf `input_amount` is consumed inside the private wrapper and does not appear in this chain-facing interface. The marginal cost consists only of the public outputs that must be published. Each authorization contributes two exit slots, each comprising an exit address (32 bytes) and a quantized amount (8 bytes, one field element), plus one nullifier (32 bytes): 112 bytes per authorization, or **56 bytes per settled output**. For comparison, an ML-DSA-87 signature alone is 4,627 bytes—over $80\times$ larger—and a full post-quantum transaction with its 2,592-byte public key would exceed 7 KB, a $>125\times$ reduction in per-output data, while providing privacy guarantees that signatures cannot offer.

9 Related Work

Wormholes and Privacy Pools: EIP-7503 proposes proof-of-burn reminting via secret-derived addresses [KBK+23]. Privacy Pools adds compliance-aware privacy sets [BIN+23]. Our work extends this line with recursive proofs that authenticate deposits via a consensus-maintained ZK tree.

Shielded Systems: Zcash provides strong privacy through shielded notes; Tornado Cash popularized mixer-style privacy on Ethereum [HBH+23, PSS19]. Our approach offers lighter-weight private settlement over visible transfer events, authenticated via a consensus-maintained ZK tree rather than encrypted note commitments.

Recursive Proofs: Halo and Plonky2 are standard recursive proof systems [BGH19, Pol22]. Plonky3 provides Poseidon2 primitives with a separate recursion track [Plo24]. We use Plonky2 with Poseidon2 for recursive composition.

Hash-Based Authorization: The Ziggy signature scheme [Sta24] uses a ZK-STARK to prove knowledge of a hash preimage, replacing lattice-based signatures with hash-based proofs.

Post-Quantum Signatures: Adoption of post-quantum signatures like ML-DSA and SLH-DSA raises questions about communication and verification cost [NIST24]. Our approach compresses authorization into ZK proofs rather than publishing large signatures.

10 Conclusion

We presented a hash-based authorization scheme where users commit funds to addresses derived from secret preimages and later claim them by proving knowledge of the preimage in zero knowledge. This approach sidesteps post-quantum signature verification entirely, requiring only hash evaluations and membership proofs inside the circuit. Recursive aggregation via Plonky2/FRI amortizes verification cost, while first-layer dummy padding and aggregate value enforcement reduce balance-linking information. We formalized the

withdrawal and private-batch relations, proved six security properties via game-based definitions, showed that computing anonymity set size is $\#P$ -hard, and demonstrated practical performance on commodity hardware.

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